

## Exercise 2 – Number System (Base form)

Example:

Decimal number system (radix or base = 10):

$$(10)_{10} \Rightarrow 1 \times 10^1 + 0 \times 10^0$$

Octal Number System:

$$(10)_8 \Rightarrow 1 \times 8^1 + 0 \times 8^0$$

Binary number system (radix or base = 2)

$$(10)_2 \Rightarrow 1 \times 2^1 + 0 \times 2^0$$

Hexa-decimal number system:

$$(0A)_{16} \Rightarrow 0 \times 16^1 + 10 \times 16^0$$

| Decimal System | Binary System | Octal System | Hexadecimal System |
|----------------|---------------|--------------|--------------------|
| 0              | 00000000      | 000          | 0                  |
| 1              | 00000001      | 001          | 1                  |
| 2              | 00000010      | 002          | 2                  |
| 3              | 00000011      | 003          | 3                  |
| 4              | 00000100      | 004          | 4                  |
| 5              | 00000101      | 005          | 5                  |
| 6              | 00000110      | 006          | 6                  |
| 7              | 00000111      | 007          | 7                  |
| 8              | 00001000      | 010          | 8                  |
| 9              | 00001001      | 011          | 9                  |
| 10             | 00001010      | 012          | A                  |
| 11             | 00001011      | 013          | B                  |
| 12             | 00001100      | 014          | C                  |
| 13             | 00001101      | 015          | D                  |
| 14             | 00001110      | 016          | E                  |
| 15             | 00001111      | 017          | F                  |

### Question 1:

Write given numbers in their base form:

$(245)_{10}$        $(142)_8$        $(101)_2$        $(74)_8$   
 $(1101)_2$        $(12)_8$        $(FF)_{16}$        $(111)_2$   
 $(777)_8$        $(11011)_8$        $(10001)_2$        $(1D0)_{16}$

### Question 2:

Convert above numbers to decimal number system using the base form notation.

(Note any number from any number system can be converted to decimal number by simply writing it in the base form notation and calculating the answer)